Optimal Reconfiguration of Systems in Aeronautics

Lecture 01: “Aircraft modeling”

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2024-25

Outline

1 Aircraft Model for the Longitudinal Motion

Objective I

This lecture provides the main lines for modeling the longitudinal motion of an aircraft.

Outline

1. Aircraft Model for the Longitudinal Motion
   - Introduction
   - Nonlinear Equations of Aircraft Motion
   - Linear Model of Aircraft Dynamics
   - Short-Period Mode
   - Phugoid Mode

Mathematical formalism I

The evolution of most engineering systems can be characterized in general form by using differential equations as

$$\dot{x}(t) = g\left[x(t), u(t), d(t), f(t)\right] \ ,$$
$$y(t) = h\left[x(t), u(t), d(t), f(t)\right] \ .$$

Still, during the initial modeling phase, there is no concrete integration of disturbances, $d(t)$, and of the term denoting the influence of faults, $f(t)$; hence, the disturbance- and fault- free nominal evolution of the system is given by

$$\dot{x}(t) = g\left[x(t), u_n(t)\right] \ , \tag{1a}$$
$$y(t) = h\left[x(t), u_n(t)\right] \ , \tag{1b}$$

where (1a) establishes the relationship between the external control inputs, $u_n(t)$, the internal states of the system, $x(t)$, and their dynamics, $\dot{x}(t)$.

In what follows, consider that all the states are measured by fault-free sensors; for simplicity, the output vector, $y(t)$, is taken identical with the state vector, $x(t)$ ((1b) makes the evolution of system dynamics available through measurement outputs, $y(t)$); hence, (1b) is not carried along in the sequel, that is the main concern rests on achieving the general representation of (1a) for the longitudinal motion of an aircraft.

Basic assumptions I

Before developing any mathematical equations of motion, a series of simplifying hypotheses must be stated:

i. The motion of the aircraft can be fully decoupled along the existing plane of symmetry into the longitudinal and lateral / directional non-interacting motions; there are no cross-couplings between the two motions in the nominal case.

ii. The aircraft is considered a rigid dense body with negligible effects of fuel sloshing or passenger wandering; moreover, the effect of spinning rotors is not taken into consideration, which is the case when symmetrical engines have opposite rotation.

iii. The aircraft moves through constant standard atmosphere and there is no wind.

Reference frames I

To develop the aircraft equations of motion, several coordinate systems employing right-handed Cartesian axes are generally used; precisely, the principal systems are the Earth-axis, F_E, and the Body-axis reference frames, F_B, respectively (other useful systems may be defined, namely Wind-axis, F_W, or Stability-axis, F_S, as well).

The principal reference frames are constructed w.r.t. the following rules:

1. Earth-axis system (F_E - $O_E x_E y_E z_E$): Axes x_E and y_E lie in the geometric plane of the Earth, which is considered flat and stationary in the inertial space; x_E-axis points North and y_E-axis points East, while z_E-axis points down toward the center of the Earth; the origin of the system, O_E, is taken at a convenient fixed point on the Earth-plane, so-called the "observer" point.

2. Body-axis system (F_B - $O_B x_B y_B z_B$): Axis x_B points out of the nose of the aircraft and is coincident with the longitudinal axis of this one, y_B-axis is directed out of the right wing of the aircraft, respectively z_B-axis is perpendicular to both x_B- and y_B- axes and is directed downward; the origin of the system, O_B, is taken to be the center of gravity (CoG) of the aircraft, the same point with the center of mass in atmosphere with uniform gravity.

Reference frames II

Usually, in order to better visualizing the displacement of the aircraft "position" regarding that of the observer, the Earth-fixed system F_E encounters a parallel translation until O_E becomes identical with CoG; this new system is called Vehicle-carried reference frame, F_V, in which $O_V \equiv O_B$ and the corresponding axes of each system are parallel, that is $x_V \parallel x_E$, $y_V \parallel y_E$, $z_V \parallel z_E$. [1]

In the sequel, the difference between the components of physical vectors expressing, e.g., position, $\overline{X}$, velocity, $\overline{V}$, acceleration, $\overline{a}$, force $\overline{F}$, or momentum $\overline{M}$, is made by attaching the corresponding subscript identifying the reference frame in which the physical variables are measured, i.e., either $(.)_V$ for variables in the Vehicle-carried system F_V or $(.)_B$ for variables in the Body-fixed system F_B, respectively.

[1] The nomenclature and terminology used is w.r.t. the definitions provided by the Group for Aeronautical Research and Technology in EURope (GARTEUR) respectively the Flight Mechanics Action Group 08 (FM AG-08) for the Research Civil Aircraft Model (RCAM).

Preliminaries I

In the following, the description of the aircraft motion is given for its two components, namely the translational and the rotational motions, as they are derived from the laws of classical physics.

The general results stated by the Newton second law and the Euler equation are extended for the aircraft, making use of the "super-particle" and "transport" theorems, under the previous assumptions of compactness and rigidity of the vehicle.

Translational motion I

The translational motion of the aircraft in the Body-axis system of coordinates, F_B, is derived from the force vector equation as follows

$$\overline{F} = m(\overline{a}_B + \overline{\Omega} \times \overline{V}_B)\,, \tag{2}$$

where the vector

$$\overline{F} = \begin{bmatrix} F_x & F_y & F_z \end{bmatrix}^\top$$

represents the total external force due to engines, aerodynamics and gravity, and m is the aircraft mass; notation $(.)^\top$ stands for transposition.

Forth,

$$\overline{V}_B = \begin{bmatrix} U_B & V_B & W_B \end{bmatrix}^\top \text{ and}$$
$$\overline{\Omega} = \begin{bmatrix} P & Q & R \end{bmatrix}^\top,$$

where $\overline{V}_B$ and $\overline{\Omega}$ are the inertial translational and rotational velocity vectors, respectively; the rotational velocity $\overline{\Omega}$ is comprised by the angular velocities P, Q, and R, denoting the roll, pitch, and yaw rotations or angular rates, expressed in the F_B system, as well.

Translational motion II

The acceleration vector $\overline{a}_B$ from (2) can be calculated as the time derivative of the inertial velocity, that is

$$\overline{a}_B = \dot{\overline{V}}_B \,;$$

moreover, the velocity $\overline{V}_B$ can be computed as the time derivative of the CoG position vector, precisely

$$\overline{V}_B = \dot{\overline{X}}_B \,;$$

however, the position of CoG is usually expressed as an Earth-fixed variable and therefore a coordinate transformation appears necessary to complete the velocity computation.

Coordinate transformation I

To describe the angular orientation of the aircraft, the vector of Euler angles

$$\Xi = \begin{bmatrix} \Phi & \Theta & \Psi \end{bmatrix}^\top$$

is introduced; provided the series of Euler angular rotations Φ, Θ, and Ψ, about the rolling, pitching, and yawing axes, the alignment of F_V with the F_B frame becomes therefore possible.

Precisely, the complete Euler transformation between F_V to F_B is computed based on the non-commutative multiplication of the direction cosine matrices corresponding to the appropriate rotation axes, i.e., $R_1(\Phi)$, $R_2(\Theta)$, and $R_3(\Psi)$, that is

$$R_{BV} = R_1(\Phi)R_2(\Theta)R_3(\Psi)\,;$$

(this transformation is unique, having one singularity point at $\Theta = \pm\frac{\pi}{2}$).

Coordinate transformation II

Viceversa, the inverse transformation from F_B to F_V can be calculated as

$$\begin{aligned} R_{VB} =& R_3(-\Psi)R_2(-\Theta)R_1(-\Phi) \\ =& \begin{bmatrix} c(\Theta)c(\Psi) & s(\Phi)s(\Theta)c(\Psi) - c(\Phi)s(\Psi) & c(\Phi)s(\Theta)c(\Psi) + s(\Phi)s(\Psi) \\ c(\Theta)s(\Psi) & s(\Phi)s(\Theta)s(\Psi) + c(\Phi)c(\Psi) & c(\Phi)s(\Theta)s(\Psi) - s(\Phi)c(\Psi) \\ -s(\Theta) & s(\Phi)c(\Theta) & c(\Phi)c(\Theta) \end{bmatrix}, \end{aligned}$$

where $s(.)$ and $c(.)$ stand for the standard trigonometric functions $\sin$ and $\cos$, respectively.

In this respect, the variation in the position of the aircraft CoG, i.e.,

$$\overline{X}_V = \begin{bmatrix} X_V & Y_V & Z_V \end{bmatrix}^\top$$

may be calculated from Body-fixed coordinates using the inverse transformation of the inertial velocity vector V_B as follows

$$\dot{\overline{X}}_V = \overline{V}_V = R_{VB}\overline{V}_B\,. \tag{3}$$

Rotational motion I

The rotational motion of the aircraft in the Body-axis system of coordinates, F_B, is derived from the moment vector equation as follows

$$\overline{M} = I\dot{\overline{\Omega}} + \overline{\Omega} \times I\overline{\Omega}\,, \tag{4}$$

where the vector

$$\overline{M} = \begin{bmatrix} L & M & N \end{bmatrix}^\top$$

represents the total external moment due to engines and aerodynamics and $\dot{\overline{\Omega}}$ is the inertial rotational acceleration expressed in the F_B frame, as well, while $I \in \mathbb{R}^{3\times 3}$ stands for the inertia tensor matrix, symmetric about the $O_B x_B z_B$ plane.

Rotational motion II

Moreover, the following relation between the rotational velocity, $\overline{\Omega}$, and the vector of Euler angles, $\overline{\Xi}$, exists

$$\dot{\overline{\Xi}} = R_{\Xi\Omega}\overline{\Omega}\,, \tag{5}$$

with the transformation matrix $R_{\Xi\Omega}$ like

$$R_{\Xi\Omega} = \begin{bmatrix} 1 & \sin\Phi\cos\Theta & \cos\Phi\tan\Theta \\ 0 & \cos\Phi & -\sin\Phi \\ 0 & \sin\Phi\sec\Theta & \cos\Phi\sec\Theta \end{bmatrix}.$$

General model I

This far, the system composed by the vectorial relations (2), (3), (4), (5), that is

$$\dot{\overline{X}}_V = R_{VB}\overline{V}_B\,, \tag{6}$$

$$\dot{\overline{\Xi}} = R_{\Xi\Omega}\overline{\Omega}\,, \tag{7}$$

$$\overline{F} = m(\overline{a}_B + \overline{\Omega}\times\overline{V}_B)\,, \tag{8}$$

$$\overline{M} = I\dot{\overline{\Omega}} + \overline{\Omega}\times I\overline{\Omega}\,, \tag{9}$$

stands for the complete description of the aircraft motion; however, the entire development is nonlinear and quite inappropriate to designing efficiently an automatic linear control law.

Airframes and motions I

To simplify the general nonlinear representation of the aircraft motion, the objective is to obtain the linear form of (6).

In this respect, the physical characteristics of the airframe system and the type of aircraft motion must be considered.

The airframe systems differentiate in the following classes:

- Class *I*: general aviation aircrafts;
- Class *II*: medium weight aircrafts;
- Class *III*: transport aircrafts;
- Class *IV*: high maneuverability aircrafts.

Furthermore, the aircraft motion classifies as follows:

- Class 1: straight and un-accelerated motion, namely level (cruise), ascent (climb), or descent (dive) flight;
- Class 2: accelerated motions, that is take-off and landing;
- Class 3: hovering flight.

Linear model I

This far, particularly for an airframe system and a specific aircraft motion, linearization may be accomplished in the small-disturbance case by perturbing the absolute values of the linear and angular velocities, and of the forces and moments, respectively, that is

$$\begin{aligned}\overline{V}_B &= \overline{V}_{B_0} + \Delta\overline{V}_B\,,\\ \overline{\Xi} &= \overline{\Xi}_0 + \Delta\overline{\Xi}\,,\\ \overline{F} &= \overline{F}_0 + \Delta\overline{F}\,,\\ \overline{M} &= \overline{M}_0 + \Delta\overline{M}\,;\end{aligned}$$

then, by developing the Taylor series expansion around the equilibrium point (steady-state trimmed values) and retaining only the first-order terms of the derived equations.

The resulting system of equations in the nominal disturbance- and fault- free operation becomes the linear state-space representation

$$\dot{x}(t) = Ax(t) + Bu_n(t)\,. \tag{10}$$

Linear model II

Insofar, due to the assumed plane of symmetry $O_B x_B z_B$, at the component-wise level, the force, moment, and linear and angular velocity vectors can be split, thus providing the longitudinal and lateral / directional subspaces of state variables.

It is also the case of the control vector $u_n(t)$, which separates exclusively the influence of the elevator deflection, namely δ_e, for the longitude, and that of the aileron and rudder deflections, namely δ_a and δ_r, for the latitude / direction, respectively.

Still, the main concern rests on the longitudinal motion analysis; hence, the standard state and control vectors are [2]

$$x_{LON} = \begin{bmatrix} u & w & q & \theta \end{bmatrix}^\top \text{ and}$$
$$u_n = \begin{bmatrix} \delta_e \end{bmatrix},$$

where u (ft/sec) and w (ft/sec) represent the inertial velocities in the x- and z-directions of F_B reference frame; also, q (rad/sec) and θ (rad) represent the pitch rate and pitch angle, respectively.

The control input δ_e (rad) is the elevator deflection.

Linear model III

Then, the longitudinal linear dynamics (10) can be expressed by the state and control matrices as follows

$$A_{LON} = \begin{bmatrix} \frac{X_u}{m} & \frac{X_w}{m} & \cdots & \frac{X_q}{m} & -g\cos\theta_0 \\ \frac{Z_u}{m-Z_{\dot{w}}} & \frac{Z_w}{m-Z_{\dot{w}}} & \cdots & \frac{Z_q+mu_0}{m-Z_{\dot{w}}} & -\frac{mg\sin\theta_0}{m-Z_{\dot{w}}} \\ \frac{1}{I_y}\left(M_u + Z_u\frac{M_{\dot{w}}}{m-Z_{\dot{w}}}\right) & \frac{1}{I_y}\left(M_w + Z_w\frac{M_{\dot{w}}}{m-Z_{\dot{w}}}\right) & \cdots & \frac{1}{I_y}\left[M_q + (Z_q+mu_0)\frac{M_{\dot{w}}}{m-Z_{\dot{w}}}\right] & -\frac{1}{I_y}(mg\sin\theta_0)\frac{M_{\dot{w}}}{m-Z_{\dot{w}}} \\ 0 & 0 & \cdots & 1 & 0 \end{bmatrix}, \tag{11a}$$

$$B_{LON} = \begin{bmatrix} \frac{X_{\delta_e}}{m} & \frac{Z_{\delta_e}}{m-Z_{\dot{w}}} & \frac{1}{I_y}\left(M_{\delta_e} + Z_{\delta_e}\frac{M_{\dot{w}}}{m-Z_{\dot{w}}}\right) & 0 \end{bmatrix}^\top; \tag{11b}$$

(if the throttle is also used for longitudinal control, then the appropriate column provided by the influence of δ_{th} is to be added in the control matrix B, as well).

Linear model IV

Note therefore that what matters in obtaining matrices A and B in (11) is the aircraft mass, m, the moment of inertia about the pitch axis, I_y, altogether with X_{var}, Z_{var}, and M_{var}, for $var \in \{u, q, w, \dot{w}, \delta_e\}$, namely the stability derivatives of the aerodynamic forces and moments w.r.t. the state-variables.

Their values may be obtained through wind-tunnel experiments or runs of Computational Fluid Dynamics (CFD) codes; they are usually tabulated for different Mach numbers at different flight conditions.

Still, the analysis of system matrix (11a) reveals two pairs of complex-conjugate eigenvalues and thus two natural modes of evolution, namely the short-period and long-period (phugoid) behaviors, which are detailed next.

[2] The variables denoted by small letters represent the relative variations from the small-disturbance case of the quantities used previously in absolute values and denoted by capital letters, thus keeping the same physical interpretation; the subscript $(.)_B$ has been removed since all the variables and quantities take values expressed in the Body-fixed reference frame, F_B.

Short-period mode I

This natural mode of evolution takes place at approximately constant speed and encounters variations in pitch attitude and angle-of-attack, α (rad) (the latter is an important state variable, which has approximately the same shape as the vertical inertial velocity, w), that is the reduced state-space vector may be defined as

$$x_{SP} = \begin{bmatrix} w & q \end{bmatrix}^\top ,$$

and thus the short-period dynamics

$$\dot{x}_{SP}(t) = A_{SP} x_{SP}(t) + B_{SP} u_n(t)$$

is described by the following state and control matrices

$$A_{SP} = \begin{bmatrix} \frac{Z_w}{m - Z_{\dot{w}}} & \frac{Z_q + m u_0}{m - Z_{\dot{w}}} \\ \frac{1}{I_y}\left(M_w + Z_w \frac{M_{\dot{w}}}{m - Z_{\dot{w}}}\right) & \frac{1}{I_y}\left[M_q + (Z_q + m u_0)\frac{M_{\dot{w}}}{m - Z_{\dot{w}}}\right] \end{bmatrix} \text{ and}$$

$$B_{SP} = \begin{bmatrix} \frac{Z_{\delta_e}}{m - Z_{\dot{w}}} \\ \frac{1}{I_y}\left(M_{\delta_e} + Z_{\delta_e} \frac{M_{\dot{w}}}{m - Z_{\dot{w}}}\right) \end{bmatrix} . \tag{12}$$

Short-period mode II

Moreover, under certain simplifying conditions, i.e.,

$$Z_{\dot{w}} \ll m \text{ and}$$
$$Z_q \ll m u_0 \,,$$

matrices from (12) reduce to

$$A_{SP} = \begin{bmatrix} \frac{Z_w}{m} & u_0 \\ \frac{1}{I_y}\left(M_w + M_{\dot{w}}\frac{Z_w}{m}\right) & \frac{1}{I_y}(M_q + M_{\dot{w}}u_0) \end{bmatrix} \text{ and}$$
$$B_{SP} = \begin{bmatrix} \frac{Z_{\delta_e}}{m} \\ \frac{1}{I_y}\left(M_{\delta_e} + M_{\dot{w}}\frac{Z_{\delta_e}}{m}\right) \end{bmatrix} . \tag{13}$$

Phugoid mode I

This natural mode of evolution encounters variations in the horizontal inertial speed and pitch attitude while the angle-of-attack remains constant, that is the reduced state-space vector may be defined as

$$x_{PH} = \begin{bmatrix} u & \theta \end{bmatrix}^\top ,$$

and the phugoid dynamics

$$\dot{x}_{PH}(t) = A_{PH} x_{PH}(t) + B_{PH} u_n(t)$$

is described by the following state and control matrices

$$A_{PH} = \begin{bmatrix} \frac{X_u}{m} + \frac{X_w}{m}\left(\frac{m u_0 M_u - Z_u M_q}{Z_w M_q - m u_0 M_w}\right) & -g \\ \frac{Z_u M_w - Z_w M_u}{Z_w M_q - m u_0 M_w} & 0 \end{bmatrix} \text{ and}$$

$$B_{PH} = \begin{bmatrix} \frac{X_{\delta_e}}{m} + \frac{X_w}{m}\left(\frac{m u_0 M_{\delta_e} - Z_{\delta_e} M_q}{Z_w M_q - m u_0 M_w}\right) \\ \frac{Z_{\delta_e} M_w - Z_w M_{\delta_e}}{Z_w M_q - m u_0 M_w} \end{bmatrix} ; \tag{14}$$

(above, $\cos\theta_0$ vanished under the assumption $\theta_0 \cong 0$).

Phugoid mode II

Moreover, under certain simplifying conditions, i.e.,

$$
\begin{aligned}
| M_u Z_w | &\ll | M_w Z_u | , \\
| M_w u_0 m | &\gg | M_q Z_w | , \text{ and} \\
| M_u X_w / M_w | &\ll X_u ,
\end{aligned}
$$

matrices from (14) reduce to

$$
\begin{aligned}
A_{PH} &= \begin{bmatrix} \frac{X_u}{m} & -g \\ -\frac{Z_u}{m u_0} & 0 \end{bmatrix} \text{ and} \\
B_{PH} &= \begin{bmatrix} \frac{X_{\delta_e} - M_{\delta_e} \frac{X_w}{M_w}}{m} \\ \frac{-Z_{\delta_e} + M_{\delta_e} \frac{Z_w}{M_w}}{m u_0} \end{bmatrix} .
\end{aligned}
\tag{15}
$$

Questions? vs. Answers!

Acknowledgments:
Marcel Staroswiecki, Dumitru Popescu,
Andrei Sperilă, Sabin Diaconescu, Teodor Rotaru,
Nicolai Christov, Cristian Oară,
Theodor Nicu.